

PilotFish XPath Evaluation Processor – Complete Reference (26R1.11 WAR)

Derived from decompiled eip.war.hs.26R1.11

August 4, 2026

At a glance (plain English)

Given XML in the transaction, this processor **runs XPath expressions** and writes the results into transaction attributes — like querying a database, but for XML elements and text. By default it uses Saxon (XPath 2+); a compatibility mode restores older XPath 1.0 behavior.

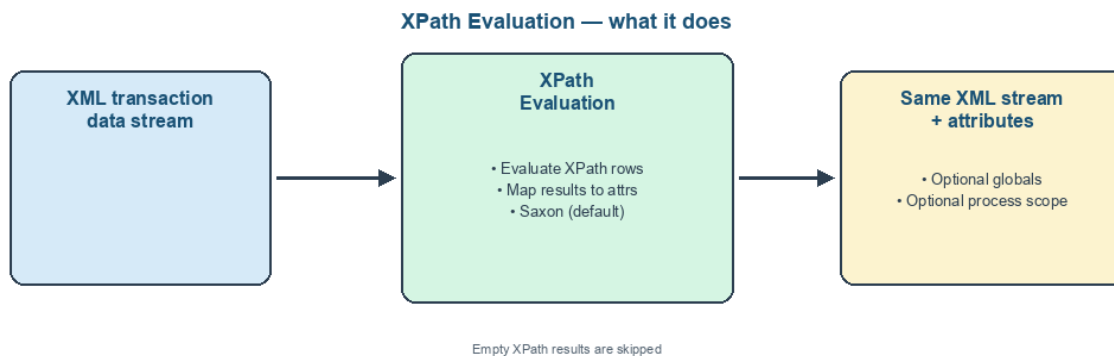


Figure 1: At a glance (plain English)

- **Evaluates** a table of Attribute / XPath pairs against the XML stream.
- **Skips** empty results (does not clear existing attributes).
- **Supports** namespace prefix mappings.
- **Can write** global bucket attributes or process-scoped attributes.

Purpose

Deep dive into **XPath Evaluation** (XPathEvaluatorProcessor) in `eip.war.hs.26R1.11`.

Provenance	Value
WAR	<code>eip.war.hs.26R1.11</code>
Module JAR	<code>xcs-modules-core-1.0.3.jar</code>
Implementation class	<code>com.pilotfish.eip.modules.other.XPathEvaluator</code>
Type string	XPath Evaluation
Help ID	<code>console.processor.xpathevaluation</code>
Categories	XPATH, ATTRIBUTES

1. Conceptual model

XML stream -> cache copy -> SaxonXPathEvaluator (default)

```
|
  For each (Attribute, XPath) row:
    value = evaluateToString(xpath)
    if non-empty -> setAttribute (transaction or process scope)
```

Optional: GlobalAttributeExpressions (Bucket, Attribute, XPath)
-> TransactionAttributes.setGlobalAttribute(...)

Stream restored from cache; XML unchanged

When **Restrict to XPath 1.0** is enabled, the legacy `XMLUtil.getNodeList` path runs instead.

2. High-level behavior

2.1 Entry (processData)

- If `XPath1Compatibility` true -> `compatibilityXPath1Process(data)`
- Else -> `xPath2Process(data)`

2.2 XPath 2+ path (XPath2Process)

1. Copy stream to default cache; restore on data.
2. `SaxonXPathEvaluator.createForDocument(cache.getInputStream())`.
3. Apply **Namespaces** table (Prefix / URI) via `addNamespace`.
4. For each **Set Transaction Attributes** row:
 - Evaluate XPath to string; if not empty (`StringUtils.isEmpty`), call `setAttribute`.
5. For each **Set Global Attributes** row (Bucket, Attribute, XPath):
 - Evaluate bucket, name, xpath through `ConfigurationManager.getValueFor`.

- If result non-empty, `TransactionAttributes.setGlobalAttribute(bucket, name, value)`.
6. Return data.

2.3 XPath 1.0 compatibility path

Uses `XMLUtil.getNodeList(data, manager, xpathList)` returning alternating nodes and attribute names; sets values from DOM text / node value / `toString()`.

2.4 Attribute scope (`setAttribute`)

- **Transaction** (default): `data.getAttributes().setAttribute(name, value)`
- **Process**: requires `ProcessOrchestrationManager.getOrCreateProcessInstance(data)` non-null; else `EIPException` mentioning `com.pilotfish.eip.process.ProcessInstanceId`.

3. Configuration reference

3.1 Set Transaction Attributes

Tag | `XPathExpressions` |
 Type | Table (Attribute / XPath) |
 Tooltip | A table of the XPath expressions and the attributes they are to be mapped to. |

3.2 Namespaces

Tag | `Namespaces` |
 Type | Table (Prefix / URI) |
 Empty keys | allowed |

3.3 Set Global Attributes

Tag | `GlobalAttributeExpressions` |
 Columns | Bucket, Attribute, XPath |
 Visible when | **Restrict to XPath 1.0** = false |
 Key repeating | allowed |

3.4 Restrict to XPath 1.0

Tag | `XPath1Compatibility` |
 Default | `false` |
 Tab | Compatibility |
 When true | shows deprecated single **Namespace** field; hides global attributes table |

3.5 Orchestration Scope (deprecated tab)

Tag | `AttributeScope` |
 Choices | `Transaction`, `Process` |
 Default | `Transaction` |

3.6 Deprecated Namespace

Tag | Namespace |

Tab | Deprecated |

Visible when | XPath 1.0 compatibility enabled |

3.7 Inherited AbstractProcessor

Tag | ExecuteProcessor |

readConfiguration calls XPathProcessorUtil.setCompatibilityMode for migration.

4. Transaction attributes

User-defined keys from the XPath table. Globals use the global attribute API. No fixed output keys except those you configure.

Runtime input for Process scope: com.pilotfish.eip.process.ProcessInstanceId.

5. Worked example

XML:

```
<Order><Id>42</Id><Partner>ACME</Partner></Order>
```

Attribute	XPath
com.acme.OrderId	/Order/Id/text()
com.acme.Partner	/Order/Partner/text()

Results: OrderId=42, Partner=ACME. XML stream unchanged.

6. Known quirks and caveats

1. **Empty XPath results skipped** — cannot use this processor to clear an attribute.
 2. **Two engines** — Saxon vs legacy XPath 1.0 differ in function support.
 3. **Process scope** fails without orchestration context.
 4. **Global rows** only in Saxon mode (compatibility false).
 5. **Stream copied to cache** — large XML duplicated in cache for duration of step.
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7. Operational recommendations

Goal	Guidance
New routes	Leave XPath 1.0 compatibility off; use Namespaces table
Legacy routes	Enable compatibility only when migrating old XPath 1 patterns
Namespaces	Declare every prefix used in XPath expressions
Globals	Reserve for cross-transaction counters/flags; avoid high churn

8. Source index

Topic	Path
XPath Evaluation	com/pilotfish/eip/modules/other/XPathEvaluatorPro
Saxon helper	com/pilotfish/eip/util/xpath/SaxonXPathEvaluator.
Module JAR	xcs-modules-core-1.0.3.jar

Deep-dive – XPath Evaluation Processor – 26R1.11 – August 2026.